COMPUTER VISION

LABSHEET 1

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ROLL NO- 23

SECTION-C

```
#1. Load and Display a Color Image
import cv2
import numpy as np
import matplotlib.pyplot as plt

# Load image using OpenCV
img = cv2.imread("image.jpg")

# Check if image is loaded
if img is None:
    print("Error: Image not found!")
else:
    print("Image loaded successfully")

# Display using OpenCV
cv2.imshow("Original Image - OpenCV", img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

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# Convert BGR to RGB for Matplotlib
img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

# Display using Matplotlib
plt.imshow(img_rgb)
plt.title("Original Image - Matplotlib")
plt.axis("off")
plt.show()

#2. Examine Image Properties
height, width, channels = img.shape

print("Width:", width)
print("Height:", height)
print("Dimensions:", img.shape)
print("Number of Channels:", channels)
print("Data Type:", img.dtype)
print("Total Pixels:", height * width)

#3. Save Image in JPEG and PNG Formats
cv2.imwrite("output.jpg", img)
cv2.imwrite("output.png", img)

print("Images saved successfully.")

#4. Convert to Grayscale, HSV and LAB
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# Grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# HSV
hsv = cv2.cvtColor(img, cv2.COLOR_BGR2HSV)

# LAB
lab = cv2.cvtColor(img, cv2.COLOR_BGR2LAB)

plt.figure(figsize=(12, 4))

plt.subplot(1, 3, 1)
plt.imshow(gray, cmap="gray")
plt.title("Grayscale")
plt.axis("off")

plt.subplot(1, 3, 2)
plt.imshow(cv2.cvtColor(hsv, cv2.COLOR_HSV2RGB))
plt.title("HSV")
plt.axis("off")

plt.subplot(1, 3, 3)
plt.imshow(cv2.cvtColor(lab, cv2.COLOR_LAB2RGB))
plt.title("LAB")
plt.axis("off")
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plt.show()

#5. Resizing
resized = cv2.resize(img, (400, 300))

plt.imshow(cv2.cvtColor(resized, cv2.COLOR_BGR2RGB))
plt.title("Resized Image")
plt.axis("off")
plt.show()

#6. Rotation

height, width = img.shape[:2]

center = (width // 2, height // 2)

matrix = cv2.getRotationMatrix2D(center, 45, 1.0)

rotated = cv2.warpAffine(img, matrix, (width, height))

plt.imshow(cv2.cvtColor(rotated, cv2.COLOR_BGR2RGB))
plt.title("Rotated Image - 45 Degrees")
plt.axis("off")
plt.show()

#7. Horizontal and Vertical Flipping
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# Horizontal flip
horizontal_flip = cv2.flip(img, 1)

# Vertical flip
vertical_flip = cv2.flip(img, 0)

plt.figure(figsize=(10, 4))

plt.subplot(1, 2, 1)
plt.imshow(cv2.cvtColor(horizontal_flip, cv2.COLOR_BGR2RGB))
plt.title("Horizontal Flip")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.imshow(cv2.cvtColor(vertical_flip, cv2.COLOR_BGR2RGB))
plt.title("Vertical Flip")
plt.axis("off")

plt.show()

#8. Generate Image Complement / Negative

negative = 255 - img

plt.imshow(cv2.cvtColor(negative, cv2.COLOR_BGR2RGB))
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plt.title("Negative Image")
plt.axis("off")
plt.show()

#9. Crop a Region of Interest (ROI)

# ROI coordinates
x1, y1 = 100, 100
x2, y2 = 300, 300

roi = img[y1:y2, x1:x2]

plt.imshow(cv2.cvtColor(roi, cv2.COLOR_BGR2RGB))
plt.title("Region of Interest (ROI)")
plt.axis("off")
plt.show()

print("ROI dimensions:", roi.shape)
print("ROI data type:", roi.dtype)
print("ROI mean intensity:", np.mean(roi))

#10. Display Original and Processed Images Together

plt.figure(figsize=(15, 10))

plt.subplot(2, 4, 1)
```

```
plt.imshow(img_rgb)
plt.title("Original")
plt.axis("off")

plt.subplot(2, 4, 2)
plt.imshow(gray, cmap="gray")
plt.title("Grayscale")
plt.axis("off")

plt.subplot(2, 4, 3)
plt.imshow(cv2.cvtColor(hsv, cv2.COLOR_HSV2RGB))
plt.title("HSV")
plt.axis("off")

plt.subplot(2, 4, 4)
plt.imshow(cv2.cvtColor(lab, cv2.COLOR_LAB2RGB))
plt.title("LAB")
plt.axis("off")

plt.subplot(2, 4, 5)
plt.imshow(cv2.cvtColor(resized, cv2.COLOR_BGR2RGB))
plt.title("Resized")
plt.axis("off")

plt.subplot(2, 4, 6)
plt.imshow(cv2.cvtColor(rotated, cv2.COLOR_BGR2RGB))
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```
plt.title("Rotated")
plt.axis("off")

plt.subplot(2, 4, 7)
plt.imshow(cv2.cvtColor(horizontal_flip, cv2.COLOR_BGR2RGB))
plt.title("Horizontal Flip")
plt.axis("off")

plt.subplot(2, 4, 8)
plt.imshow(cv2.cvtColor(negative, cv2.COLOR_BGR2RGB))
plt.title("Negative")
plt.axis("off")

plt.tight_layout()
plt.show()
```

OUTPUT-



QUESTION AND ANSWER

1. What is a digital image? Differentiate between grayscale and color images.

Answer: A digital image is a collection of pixels represented digitally, where each pixel contains intensity or color information. Grayscale image: Has only one channel, representing intensity from black to white. Color image: Usually has three channels such as Red, Green, and Blue (RGB).

2. Explain the difference between RGB, BGR, HSV, and LAB color spaces.

Answer: RGB: Represents colors using Red, Green, and Blue components. BGR: Same components as RGB but stored in Blue, Green, Red order; commonly used by OpenCV. HSV: Represents Hue, Saturation, and Value; useful for color-based segmentation. LAB: Represents Lightness (L) and color components (A and B); useful for color analysis and correction.

3. What is the purpose of converting an image to grayscale before further processing?

Answer: Grayscale conversion reduces a three-channel color image to a single intensity channel. It reduces computational complexity and is useful for operations such as edge detection, thresholding, segmentation, and feature extraction.

4. Explain the concept of image complement (negative image) and mention its practical applications.

Answer: An image complement or negative image is produced by reversing the intensity values:

$I_{negative} = 255 - I$

For example, black becomes white and white becomes black. It is used in medical imaging, document analysis, photography, and enhancing certain image details.

5. Differentiate between image resizing, cropping, and scaling.

Answer:

Operation	Meaning
Resizing	Changes the image dimensions, such as 640×480 to 320×240.
Cropping	Removes unwanted areas and keeps only a selected region.
Scaling	Enlarges or reduces the size of an image, usually by a specific scale factor.

6. What is a Region of Interest (ROI), and why is it important in computer vision?

Answer: ROI is a selected portion of an image containing the important object or area for analysis. It is important because it reduces unnecessary processing, saves computation, and allows algorithms to focus on relevant information.

7. Why is OpenCV preferred over conventional image processing libraries for computer vision applications?

Answer: OpenCV is preferred because it is open-source, fast, cross-platform, and provides many built-in computer vision functions. It supports image processing, object detection, feature extraction, video processing, and machine learning operations.

8. Explain how image resolution and pixel intensity influence image quality.

Answer: Resolution determines the amount of spatial detail in an image. Higher resolution generally provides more detail. Pixel intensity represents the brightness of a pixel. Proper intensity levels provide better contrast, while very low or high intensities can cause dark or overexposed regions.

9. Mention five real-world applications where basic image preprocessing is an essential step.

Answer: Five applications are: Medical image analysis – processing X-rays and MRI images. Face recognition – resizing and normalization of face images. Autonomous vehicles – preprocessing camera images. OCR/document analysis – grayscale and thresholding. Satellite image analysis – enhancement and segmentation.

10. How do image preprocessing techniques improve the performance of feature extraction and deep learning models?

Answer: Preprocessing improves image quality by removing noise, standardizing image size, improving contrast, and selecting relevant regions. This helps feature extraction algorithms identify useful patterns and allows deep learning models to train faster, more consistently, and often with better accuracy.